

PU-I Biology · Chapter 5

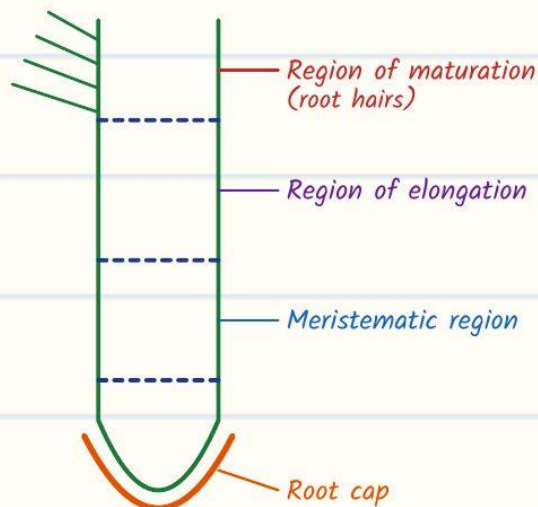
Morphology of Flowering Plants

Handwritten Notes — Narendra Sir · BAIO Learning System

1 The Root System

Def ► Root = descending part of the plant axis, develops from the **radicle**; absorbs water & minerals, anchors the plant.

- **Tap root** — primary root + lateral roots · dicots (mustard)
- **Fibrous root** — primary root short-lived, many roots from stem base · monocots (wheat)
- **Adventitious root** — from parts other than radicle (grass, Monstera, banyan)



Regions of the root tip

Functions

- absorption of water & minerals
- anchorage
- storage of reserve food
- synthesis of plant growth regulators

Modifications

- **Storage:** carrot, turnip (tap) · sweet potato (adventitious)
- **Prop roots:** banyan · **stilt roots:** maize, sugarcane
- **Pneumatophores:** Rhizophora (respiration in swamps) ★ exam

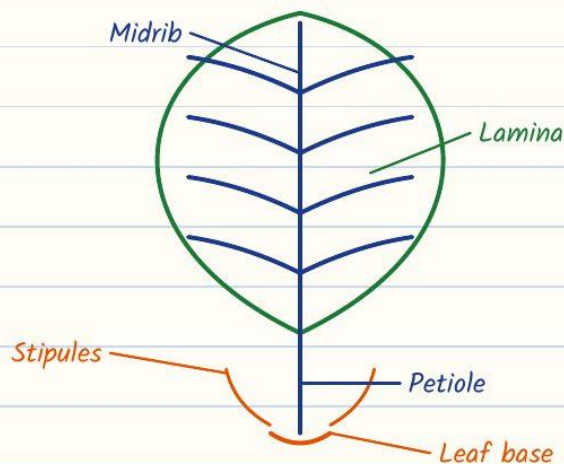
2 The Stem

Def ▶ Stem = ascending part of the axis bearing branches, leaves, flowers & fruits — develops from the **plumule**.

- Has **nodes** (where leaves arise) and **internodes** · bears terminal & axillary buds
- Functions: spreading branches, conduction, support · some store food, support, protect, propagate

Modification	Examples	Purpose
Underground	potato (tuber), ginger & turmeric (rhizome), Colocasia (corm), onion (bulb)	food storage, perennation
Stem tendrils	cucumber, pumpkin, grapevine	climbing
Thorns	Citrus, Bougainvillea	protection
Phylloclade	Opuntia (flattened) · Euphorbia (fleshy)	photosynthesis in xerophytes
Sub-aerial	runner (grass), stolon (jasmine), offset (Pistia, Eichhornia), sucker (banana, mint)	vegetative propagation

3 The Leaf



Parts of a leaf

- Lateral, flattened structure on stem; develops at the node, bears a bud in its axil
- 3 parts: **leaf base**, **petiole**, **lamina** · stipules may be present
- **Pulvinus** = swollen leaf base (legumes)

Venation

- **Reticulate** — network · dicots
- **Parallel** — veins parallel · monocots

Simple leaf — lamina entire or incised, incisions don't touch the midrib

Compound leaf — incisions reach midrib → leaflets · **pinnate** (neem) · **palmate** (silk cotton)

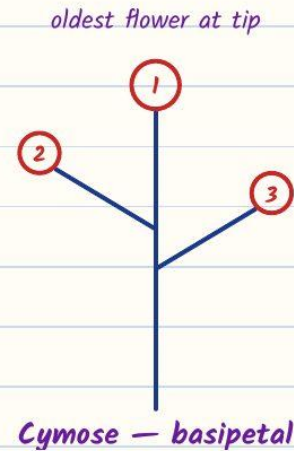
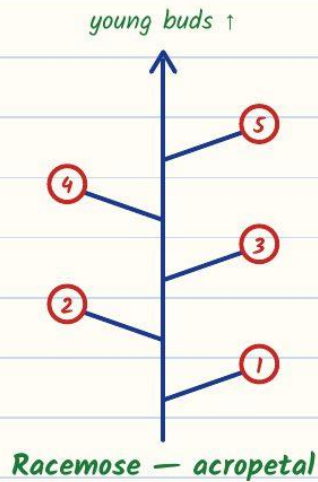
Phyllotaxy ★ exam

Type	Leaves per node	Example
Alternate	1	china rose, mustard, sunflower
Opposite	2, opposite	Calotropis, guava
Whorled	> 2	Alstonia

- **Leaf modifications**: tendrils (pea), spines (cactus), fleshy storage (onion, garlic), phyllode (Australian Acacia), pitcher (Nepenthes)

4 The Inflorescence

Def ► Arrangement of flowers on the floral axis. A flower is a **modified shoot** — the shoot apex turns into a floral meristem, internodes don't elongate.



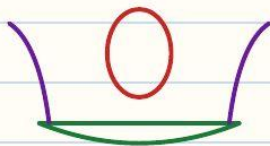
💡 **Remember:** Racemose: main axis keeps growing, flowers **acropetal** (old at base).
Cymose: main axis ends in a flower, flowers **basipetal**.

5 The Flower

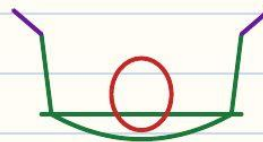
- 4 whorls on the thalamus: **calyx** (sepals) · **corolla** (petals) · **androecium** (stamens, ♂) · **gynoecium** (carpels, ♀)
- Calyx + corolla similar → **perianth** (lily)
- Symmetry: **actinomorphic** (+) (mustard, Datura, chilli) · **zygomorphic** % (pea, Cassia, gulmohur) · **asymmetric** (Canna)
- Trimerous / tetramerous / pentamerous = floral parts in multiples of 3 / 4 / 5

Position of floral parts on thalamus

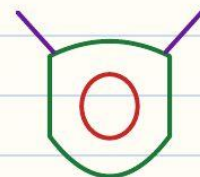
ovary = red · thalamus = green · other whorls = purple



Hypogynous
superior ovary



Perigynous
half-inferior



Epigynous
inferior ovary

Hypogynous: mustard, china rose, brinjal · Perigynous: plum, rose, peach · Epigynous: guava, cucumber, ray florets of sunflower

Aestivation (arrangement of sepals / petals in the bud)

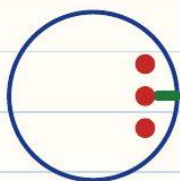
- **Valvate** — margins just touch (Calotropis) · **Twisted** — one overlaps next (china rose, cotton)

- **Imbricate** — irregular overlap (Cassia, gulmohur) · **Vexillary / papilionaceous** — standard + wings + keel (pea, bean) ★ exam

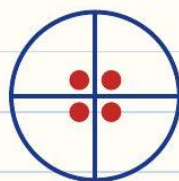
Androecium & Gynoecium

- **Stamen** = filament + anther (2-lobed, pollen sacs) · **staminode** = sterile stamen
- **Epipetalous** (brinjal) · **epiphyllous** (lily) · **monodelphous** (china rose) · **diadelphous** (pea) · **polyadelphous** (citrus)
- **Carpel** = stigma + style + ovary · **apocarpous** (free: lotus, rose) · **syncarpous** (fused: mustard, tomato)

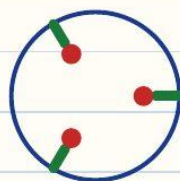
Placentation



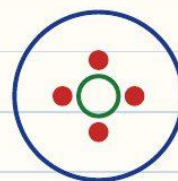
Marginal
pea



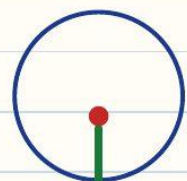
Axile
tomato, lemon



Parietal
mustard



Free central
Dianthus



Basal
sunflower

6 Fruits and Seeds

- **Fruit** = mature / ripened ovary after fertilisation · without fertilisation → **parthenocarpic fruit**
- **Pericarp** → **epicarp** (outer) · **mesocarp** (middle) · **endocarp** (inner)
- **Mango & coconut** = **drupe** (from monocarpellary superior ovary, one-seeded) · mango: fleshy edible mesocarp · coconut: fibrous mesocarp ★ exam

Dicot seed (bean, gram, pea)

seed coat (testa + tegmen) · hilum · micropyle · embryo = embryonal axis + 2 cotyledons

Monocot seed (maize)

endospermic · aleurone layer · **scutellum** (1 cotyledon) · plumule in coleoptile, radicle in coleorhiza

7 Family Solanaceae (Potato family)

Floral formula: $\oplus \quad \text{♀} \quad K_{(5)} \quad C_{(5)} \quad A_5 \quad \underline{G}_{(2)}$ ★ exam

- **Vegetative:** mostly herbs/shrubs · stem erect, cylindrical, hairy · leaves alternate, simple, reticulate
- **Inflorescence:** solitary, axillary or cymose (Solanum)
- **Flower:** bisexual, actinomorphic · calyx 5 united, **persistent** · corolla 5 united, valvate
- **Androecium:** 5 stamens, **epipetalous** · **Gynoecium:** bicarpellary, syncarpous, superior ovary, bilocular, **swollen placenta, axile**
- **Fruit:** berry or capsule · seeds many, endospermous

Use	Plants
Food	tomato, brinjal, potato
Spice	chilli
Medicine	belladonna, ashwagandha
Fumigatory	tobacco
Ornamental	petunia

Quick revision ✓

Tap (dicot) vs fibrous (monocot) · pneumatophores = Rhizophora · phylloclade = Opuntia · racemose acropetal / cymose basipetal · vexillary = pea · axile = tomato · drupe = mango, coconut · scutellum = maize · Solanaceae ⊕ ♀ K(5) C(5) A5 G(2)